

Validation of a Gamified Measure of Processing Speed

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Abstract

Performance on many cognitive tasks is underpinned by processing speed. Processing speed is poorly defined and operationalised in the literature, but critics suggest utilizing simplified tasks for its measurement. These tasks are, by nature, repetitive and boring, often long, poor ecological validity and failure to establish respondents' optimal capacity. The introduction of gamified elements to cognitive tasks already shows promise of overcoming issues relating to boredom and ecological validity, however limitations remain for the most fundamental and tedious of assessments, the simple reaction time task (SRT). The present study assesses participants' performance and experience ratings of a gamified version of the SRT, DoggoNogo, compared to a stripped-back non-gamified version. Our preregistered hypotheses were that (i) participants would find DoggoNogo more engaging, reflected by higher enjoyment ratings and lower reaction time (RT) variability, and (ii) DoggoNogo would demonstrate good convergent validity, reflected by strong correlations in RT indices of centrality and dispersion between the two tasks. Bayesian t-tests indicated extreme evidence that participants enjoyed DoggoNogo more than the non-gamified SRT, but they also presented greater RT variability in DoggoNogo. Based on participant feedback, we suspect the latter result may be confounded by the difference in perceptual load between task versions, and may not accurately reflect participant engagement. Bayesian correlations found poor convergent validity among all indices except mean and median RTs. Exploratory analyses show the promise of DoggoNogo, as reliable indices can be obtained over twice as fast as in the non-gamified task. Limitations and future directions are discussed.

Keywords: Gamification, Processing Speed, Cognitive Control, Reaction Time

Research Transparency Statement

Conflicts of interest: None. Funding: None. Artificial intelligence: No artificial intelligence assisted technologies were used in this research or the creation of this article. Ethics: This research received approval from a Psychology School Research Ethics Officer at the University of Sussex (ER/BMT26/6).

Preregistration: The hypotheses, methods and analysis plan were preregistered on the Open Science Framework (OSF; <https://doi.org/10.17605/OSF.IO/FUMQC>) prior to data collection. Any deviations or additions are explicated in text. All study materials, primary data, and analysis scripts are publicly available at <https://github.com/RealityBending/DoggoNogoValidation>.

Introduction

Processing speed (PS) is an element of cognitive control which underpins performance across many cognitive tasks. Simply, it may be described as the duration between being exposed to information and understanding it such that you can purposefully respond to it. Quick processing facilitates cognitive efficiency (e.g., a chess player rapidly evaluates options in a timed match) and personal safety; a driver breaks on seeing a car pull into the road ahead; medical staff move away from a patient on hearing ‘clear’. From a neural standpoint, changes in PS may occur because of disruption in the broader cognitive control network, described by Cole and Schneider (2007). Specifically, PS has been associated with abnormal activity in the dorsolateral prefrontal cortex (Panagiotaropoulou et al., 2019), and reduced fronto-parietal activation (Basner et al., 2013). Processing speed can be compromised temporarily, for example, by sleep deprivation (Basner et al., 2013; Bougard et al., 2016). It can also be chronically

impaired, for example, in demyelinating diseases such as multiple sclerosis ([Sivakolundu et al., 2020](#), for review), or psychiatric conditions such as schizophrenia ([Panagiotaropoulou et al., 2019](#)) and bipolar disorder ([Little et al., 2023](#)). When PS is compromised, response time (RT) increases; the chess player may blunder; the driver may crash; the medic may receive a nasty shock.

Investigations into individual differences in PS have yielded mixed results. For instance, some research suggests that females have longer PSs than males ([Jakobsen et al., 2011](#)) while others found no difference ([Woods et al., 2015](#)). A review by Roivainen ([2011](#)) found that gender differences in PS are task dependent, with females faster across Stroop and symbol matching tasks, and males faster for reaction time and tapping tasks. Consistent with this Ghisletta et al. ([2017a](#)) found that in a simple reaction time task, males were faster among children and young adults, but not older adults. A wealth of evidence supports age-related slowing ([Salthouse, 1996](#); [Talboom et al., 2021](#)). Woods et al. ([2015](#)) argues that this is primarily due to the motor demands rather than the stimulus detection facet of PS tasks, which would be consistent with research indicating no effect of age on speed when participants respond verbally ([Nebes, 1978](#)). However, a meta-analysis by Dykiert et al. ([2012](#)) found that age-related slowing was more prominent in choice- than simple-reaction tasks, suggesting an important contribution of the cognitive element. In addition to personal characteristics, PS is associated with fluid intelligence ([Sheppard & Vernon, 2008](#)), physical components of quality of life ([Jakobsen et al., 2011](#)), and instrumental activities of daily living ([Wadley et al., 2021](#)).

Despite a wealth of research, PS has been poorly defined and operationalised throughout the literature ([Rommelse et al., 2020](#)), casting doubt on the validity of inferences yielded from

use of more complex PS tasks that tap into higher cognitive functions ([Cepeda et al., 2013](#)). For instance, the ‘Coding’ and ‘Symbol search’ sub-tests of the Wechsler Intelligence Scales are posited to assess PS: These pen-and-paper tasks allow respondents 2 minutes to match as many symbols to numbers as possible according to a key (Coding), or to identify whether target symbols are present among a series of symbols (Symbol search). These tests are mixed-domain, involving short-term memory demands, fine motor-skills and visual scanning ([Rommelse et al., 2020](#)), and thus fail to isolate PS. To address this, Cepeda et al. ([2013](#)) recommended the use of simpler PS tasks. A simple reaction time task minimises demands on participants, requiring only that they respond (e.g., by pressing a button) as quickly as possible when the stimulus occurs. However, such tasks have their own limitations; they are repetitive by nature, often with pre-set RT thresholds that may allow participants to gauge the minimum necessary effort required. This approach, while perhaps sufficient to identify clear impairments, does not facilitate the accurate measure of individuals’ optimum PS capacity and therefore cannot be used to accurately account for PS in more complex cognitive tasks ([Draheim et al., 2019](#)). This may have knock-on effects for identifying where deficits exist and prevent the implementation of appropriate support. One solution, described by Draheim et al. ([2019](#)), is use of adaptive thresholds: adjusting task difficulty according to participant performance and allowing for more precise and efficient assessment.

Rouder et al. ([2023](#)) highlight the issue of low signal-to-noise ratios assessing individual differences in cognitive control tasks. They suggest a solution may be improving engagement via gamification. Gamification is being applied increasingly to cognitive tasks, with researchers taking this approach holding expectations of improvements in participant motivation, long-term

engagement, ecological validity, and suitability for target age groups or disorders (Lumsden et al., 2016). However, in their review, Lumsden et al. (2016) investigated gamified cognitive assessments and found that these studies were still limited by targeting multiple domains. The two gamified PS tasks (McPherson & Burns, 2007, 2008), did not measure simple reaction time, and both failed to disentangle PS from working memory. Lumsden et al. (2016) asserted that gamification does not improve data quality despite improving participant experience, recommending the development of single-domain gamified tasks and the incorporation of non-gamified versions for comparison.

Following these recommendations, Robison and Nguyen (2023) applied gamification to a simple RT task, investigating its effects on performance and vigilance decrement (i.e., the tendency for RTs to increase over time). Comparing their task to pre-existing data from a non-gamified psychomotor vigilance task, they found that gamification produced better performance and lower vigilance decrement when the game involved a competitive aspect, but only produced better performance and lower vigilance decrement in non-competitive games when there was a point-system.

Pedersen et al. (2023) included an additive procedure in their game *Rat Catch*, such that level 1 assessed simple reaction time, presenting the stimulus for decreasing durations, before proceeding to choice reaction time. In this way, lower level cognitive functions can be accounted for in assessments of performance as tasks become more complex. The authors asserted good reliability and convergent validity across dimensions in gamified compared to standard tasks. However, parameters were static across participants. One disadvantage of this is that a brief distraction could yield unreliable game data for a participant. To avoid this issue, participant

progression could be determined by a points system, with stimulus duration set dynamically to identify participants' optimal ability. A similar suite of gamified cognitive assessments, MindPulse, was developed by Suarez et al. (2021) to disentangle components of reaction time. While promising, the global scientific community is heavily reliant on open software, so access to these tools is limited.

Extending Robison and Nguyen (2023) and Pedersen et al. (2023), the present study compares open-source non-gamified SRT task to a gamified version that incorporates a point-system and self-competing dynamic RT threshold. We seek to validate and optimize a simple reaction time task by (1) testing the impact of gamification on PS and task engagement and (2) assessing the optimal parameter values (number of trials and ISI range) for a robust yet efficient measure.

Materials and Methods

Participants

Participants were recruited online from the University of Sussex participant pool via Sona Systems (<https://www.sona-systems.com/>) and were awarded with course credits. Fluency in English was the only requirement for participation. Participants provided informed consent at the beginning of the experiment, and reaffirmed their consent after being debriefed when asked to continue to the next screen, at which point their data would be saved to our OSF data repository via DataPipe (Leeuw, 2024). Participants could stop the experiment at any time. However, because it was hosted externally, data from the DoggoNogo (gamified) task was sent to OSF via DataPipe on completion of the game. Therefore, any participants for whom we obtained

incomplete data were deleted, with the assumption that non-completion indicated a withdrawal of consent.

Complete data¹ for 86351 trials was obtained from 315 participants. We removed participants who missed more than one-third of trials² ($N = 9$), and those with non-typical RT distributions. Typical reaction time data is positively skewed and leptokurtic ([Whelan, 2008](#)). Therefore, RT distributions were assessed for each participant and task combination by visualising distributions that either had skewness values less than 0.5 (i.e., symmetric or negatively skewed), or both kurtosis values of less than 0.5 (i.e., meso- or platykurtic) and a non-significant Shapiro-Wilk test of normality on the log-transformed RT data. All distributions that fell into this criteria ($N = 31$; 1 = DoggoNogo, 30 = SRT) appeared atypical for RT distributions and these participants were excluded. Finally, in line with recommendations by Miller ([2023](#)), data for all remaining trials that had implausibly fast RTs ($<150\text{ms}$) were removed. We preregistered exclusion criteria of trials with RTs $>1000\text{ms}$, but no such cases were present. The final sample consisted of 275 participants (Mean age = 19.6, SD = 2.3, range: [18, 51]; Gender: 81.8% women, 16.7% men, 1.45% non-binary; Education: High school, 81.45%; Bachelor, 14.55%; Master, 0.36%; Doctorate, 1.45%; Other, 2.18%) for a total of 72958 trials.

¹ Our preregistration stated that participants for whom DoggoNogo data was missing would be removed ($N = 0$). We also removed participants who were missing SRT data but not DoggoNogo data ($N = 7$), as this meant that the study was exited before completion, indicating withdrawal of consent.

² This was not part of our preregistered exclusion criteria. However, the simple RT task had a constant RT threshold of 600ms and visual simple reaction time averages between 217.9 and 239.1ms (SD range: 19.5-28.1ms) ([Woods et al., 2015](#)). Therefore, we conservatively expect that any participant who did not successfully respond to two-thirds of trials was likely not sufficiently engaged with the task, or was disproportionately affected by hardware or software-induced lag, rendering their data unreliable.

Measures and Procedure

After providing consent and demographic information, participants completed two computer-based simple reaction time tasks: a gamified version (DoggoNogo) and a non-gamified version (SRT). The order of tasks was counterbalanced across participants. Both tasks involved three blocks with opportunities for breaks in-between. Participants responded using the down arrow on their keyboard.

After each of the tasks participants responded to three short questions about their experience. To assess task enjoyment, participants rated the statement “How much did you enjoy the previous task?” on an 8-point likert scale from 0 = “boring” to 7 = “fun”. They then provided an estimation of task duration in minutes, “Without checking the time, how long do you think you spent doing the previous task?”, and finally indicated their willingness to repeat the task by rating the statement: “How would you feel if you had to do the same task again one more time?” on a 9-point likert scale from -4 = “very annoyed” to 4 “very happy”.

After the first of the two reaction time tasks, participants completed a 9-item Raven’s Progressive Matrices (RPM, [Bilker et al., 2012](#)). This task is a measure of fluid intelligence which has robust associations with PS ([Sheppard & Vernon, 2008](#), for review). The RPM was implemented for the purpose of assessing convergent validity after data collection began, and therefore was not preregistered and was only completed by 138 (48.59%) of the final sample. Its use falls outside of the scope of this paper and is therefore not reported further.

At the end of the experiment, participants had an opportunity to provide written feedback and rated their overall enjoyment of the experiment on a 5-point Likert scale.

DoggoNogo

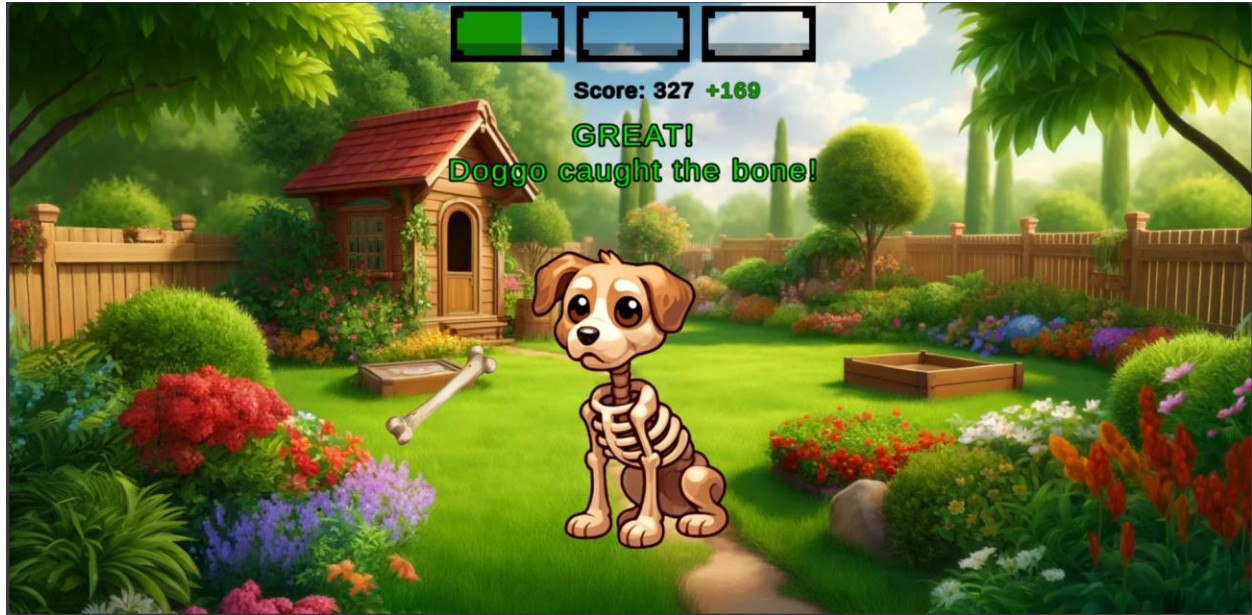
DoggoNogo was written in Unity (version 2022.3.13f1; <https://unity.com/>)³. The task is preceded by a short visual narrative introducing Doggo's background and the aim of saving the malnourished dog. In the game, participants must catch bones for Doggo by pressing the down arrow as quickly as possible whenever a bone appears on screen. Game performance determines Doggo's health, and this is reflected by the health bars throughout gameplay as well as changes in character appearance in each block.

The game is set in a garden with Doggo centered (Figure 1). Doggo remains static throughout, but makes a small 'jump' whenever the down arrow is pressed. The bone is static; size varied (30-70% of the original size) as did position on the screen (randomized whilst avoiding overlap with Doggo or the health bars) and rotation (a random value between 0 and 360 degrees).

³ The development version of DoggoNogo can be accessed on GitHub at <https://github.com/SussexPsychologySoftware/DoggoNogo>

Figure 1

DoggoNogo example display



The number of trials per block was dependent on participant performance; participants had to obtain 2000 points in each block before proceeding to the next block or completing the game, and could earn 100-200 points per trial.

Bones appear following interstimulus intervals (ISIs) of durations randomly obtained on each trial from a uniform distribution between 1000ms and 4000ms.

Points were obtained by catching bones by pressing the down arrow on the keyboard, and both success and number of points depend on participant RT relative to their cumulative median RT. Premature responses received -100 points (but minimum total score is limited to zero), RTs longer than double the participants cumulative median were 'missed' and yielded 0 points and RTs longer than participants cumulative median were 'slow' and obtained 0 points. RTs shorter than participant cumulative medians were 'fast' and yielded a score based on the following

(equation 1): RT is clamped to the range between 150ms and double the participant's cumulative median RT ($2RT_{\text{Median}}$), normalised and subtracted from 1. The result is multiplied by the score range (i.e., $200 - 100 = 100$) and added to the minimum score (100). This is the participants' final score, unless it exceeds 200, in which case the participant obtains 200 points. This scoring system introduces an implicit competition element and ensures points are received for approximately half of all trials. At the beginning of the experiment, the RT bounds were set to $RT_{\text{min}} = 150\text{ms}$ and $2RT_{\text{Median}} = 600\text{ms}$, and our score limits set to $[100, 200]$ throughout.

$$\begin{aligned} \text{Score} &= \text{Score}_{\text{min}} + \left(1 - \frac{RT_{\text{Clamped}} - RT_{\text{min}}}{2RT_{\text{Median}} - RT_{\text{min}}}\right) \times (\text{Score}_{\text{min}} - \text{Score}_{\text{max}}) \\ \therefore \text{Score} &= 100 + \left(1 - \frac{RT_{\text{Clamped}} - 150}{2RT_{\text{Median}} - 150}\right) \times 100 \end{aligned} \quad (1)$$

DoggoNogo was completed after between 86 and 219 trials ($M = 119.76$, $SD = 17.77$) and for a duration (including time-unlimited breaks) of between 6.03 and 88.45 minutes ($M = 8.84$, $SD = 5.3$).

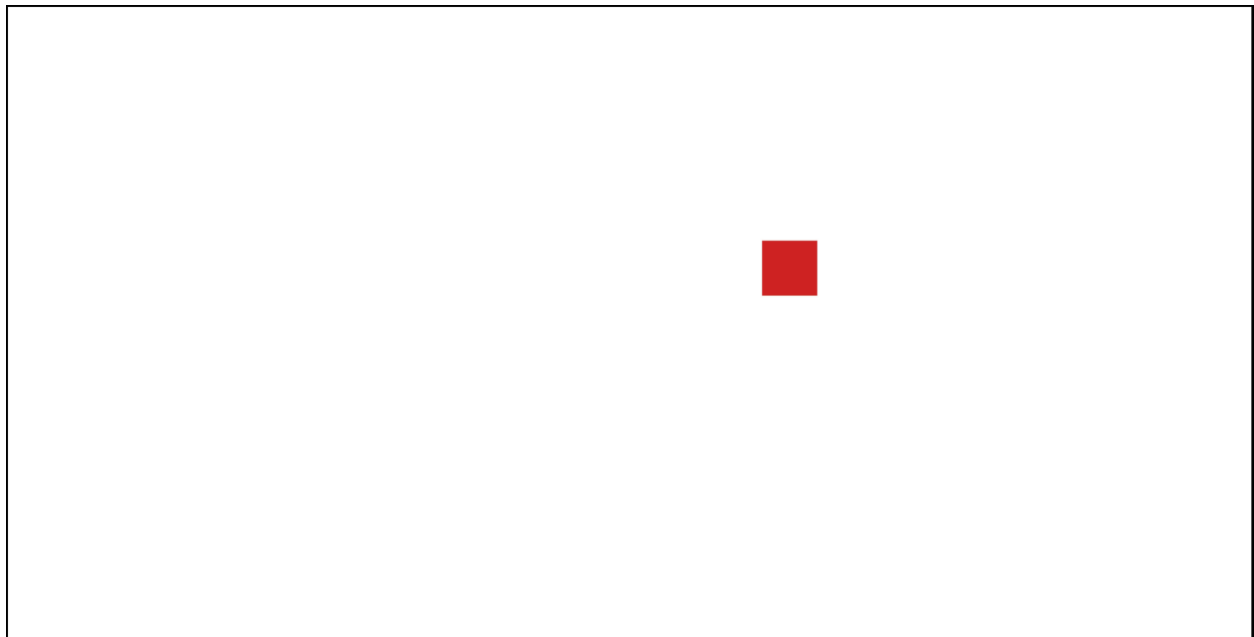
SRT

The SRT was written using jsPsych version 8.0.2 (Leeuw et al., 2023). Participants were informed that during the task they would see a fixation cross (+) followed by a red square. They were instructed to press the down arrow on their keyboard as fast as possible when the red square appears (Figure 2). The display background was white and the fixation cross appeared as a black plus (+) symbol in the middle of the screen. The fixation cross was visible for the ISI duration; a value randomly selected for each trial from a uniform distribution ranging 1000-4000(ms). Premature responses elicited a message ("Too fast! Wait until the red square appears") and reset

the trial with a new ISI value - thus preventing participants from skipping through the experiment. At the end of the ISI, the fixation cross was replaced with a red square (target stimulus) which appeared in a random location (vertical and horizontal offsets from centre each randomly selected within $\pm 90\%$ of the screen width and height) and size (randomly selected from a discrete uniform distribution from 50px to 200px, in steps of 10px) on the screen. If participants did not respond to the target stimulus within 600ms, their RT was recorded as null and they progressed to the next trial. There were three blocks of 50 trials each (150 trials in total), separated by breaks of choice duration. Participants successfully responded to between 101 and 150 trials ($M = 145.54$, $SD = 6.19$), and completed the task in between 6.96 and 30.86 minutes ($M = 8.87$, $SD = 2.76$).

Figure 2

Non-gamified Simple RT task example display



Analytic Procedure

All preprocessing and analyses were conducted in R version 4.4.1 ([R Core Team, 2024](#)).

Preprocessing

The preprocessing and analysis files can be found in our GitHub repository. Raw data was in .csv format for the SRT, and JSON format for DoggoNogo. Preprocessing therefore involved restructuring the data as well as calculating times and durations, removing Sona survey IDs, and removing participant data where withdrawal of consent was implied. Data cleaning procedures included the visualisation of reaction time distributions and filtering data according to preregistered criteria.

Confirmatory analyses

Our preregistered inference criteria were Bayes Factors exceeding 3. Credible intervals were calculated at the 95% level using the highest density interval (HDI). The method for calculating CIs was not preregistered, however, this method was selected as it captures the most probable values, including the maximum a posteriori (i.e., single most probable value) which an equal tailed interval method is not guaranteed to capture.

Bayesian correlations were conducted using *correlationBF()* from the *BayesFactor* package ([Morey & Rouder, 2024](#)), using a shifted and scaled $Beta\left(\frac{1}{3}, \frac{1}{3}\right)$ (“medium”) prior distribution, for the hypothesis that there exists a non-zero relationship between tasks across RT indices (H_1). The posterior distribution was approximated using Markov Chain Monte Carlo (MCMC; 1000 iterations), and trace plots (Appendix A) appeared noisy but flat, suggesting independent sample draws and no convergence issues. The region of practical equivalence (ROPE) was set such that values in the region of $[-1; 0.5]$ are considered negligible, in line with

our hypothesis that a strong correlation exists between tasks; the upper value is based on Cohen (1988) threshold for a ‘strong’ correlation, set as the minimum value necessary to indicate convergent validity. Correlations were calculated between RT indices (i.e., mean, median, mode, SD, MAD, IQR) for the two tasks to assess the validity of our gamified task. Because of the typical skew in RT distributions, assessing all of these indices will provide a more comprehensive picture of the differences in performance between tasks.

Bayesian t-tests were conducted using *ttestBF* from the BayesFactor package (Morey & Rouder, 2024) using default priors ($\sqrt{2}/2$). These were conducted to assess differences in participant engagement between tasks via enjoyment ratings and RT variability (SDs).

Exploratory analyses

To assess whether RT varies according to gender or handedness, we conducted bayesian hierarchical regressions using *brm()* from the *brms* package Bürkner (2017) and default priors. We first modelled main effects of Gender (or Handedness) on each of the RT indices with random effects for participant, and then added interaction effects between Gender (or Handedness) and task version. We extracted model parameters using *parameters()* from the *parameters* package (Lüdtke et al., 2020), performance indices using *performance()* from the *performance* package (Lüdtke et al., 2021), Bayes factors using *bayesfactor()* from the *bayestestR* package Makowski, Ben-Shachar, and Lüdtke (2019), and marginal means using *estimate_means()* from the *modelbased* package (Makowski et al., 2020).

Finally, with the aim of optimizing the robustness and efficiency of DoggoNogo, we assessed the optimal values for ISI and number of trials by modelling the effect of ISI on RT,

and assessing the convergence of cumulative RT indices between tasks as a function of the number of trials.

Results

Confirmatory

The relationship between tasks for each index of RT can be seen in Figure 3.

Figure 3

Scatter plots showing RT indices' relationships between tasks.

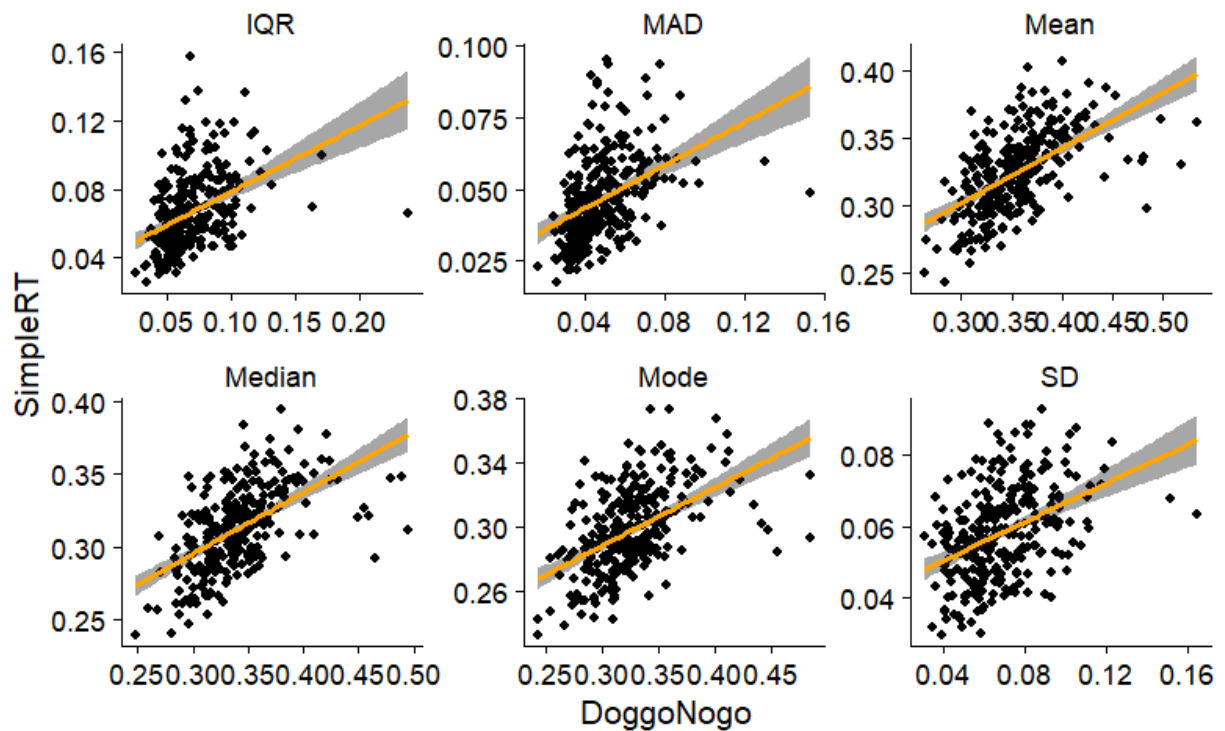


Table 1*Bayesian correlation estimates between tasks' RT indices*

	ρ_{Median}	ρ_{MAD}	95% CI (HDI)	$BF_{10} (>0)$	pd	$P_{in\ rope}$	$BF_{10} (>ROPE)$
Mean	0.56	.04	[0.47, 0.64]	>1Q	1	6.41%	4.85
Median	0.55	.04	[0.47, 0.63]	>1Q	1	6.94%	4.46
Mode	0.50	.04	[0.42, 0.59]	>1Q	1	47.5%	0.46
SD	0.41	.05	[0.32, 0.50]	12.3B	1	100%	0.007
MAD	0.40	.05	[0.30, 0.49]	4.3B	1	100%	0.009
IQR	0.41	.05	[0.31, 0.50]	15.6B	1	100%	0.012

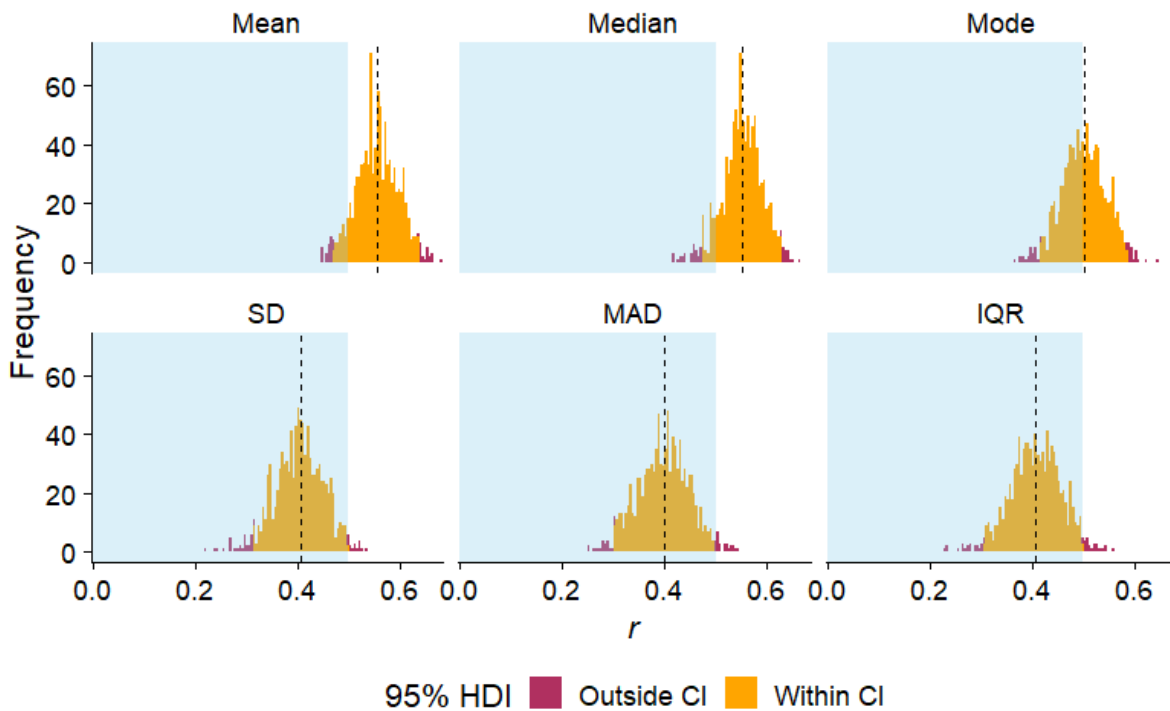
Note. MAD = median absolute deviation. 95% CI (HDI) = Credible interval (highest density interval). pd = probability of direction. BF_{10} = Bayes factor (H_1/H_0). ROPE = Region of Practical Equivalence. B = Billion. Q = Quadrillion.

Results indicate very strong evidence of a positive correlation between tasks for each index of RT, $BF_{10} > 4.3\ Billion$, and a probability of direction of 100% for being positive across all indices. However, with median values of the correlation coefficient posteriors ranging from 0.40 to 0.56, there are considerable overlaps between the ROPE and 95% HDIs Figure 4. As can be seen in Table 1, the HDI of all indices of dispersion (i.e., SD, MAD, IQR) fell entirely within the ROPE region, and those for indices of centrality fell between 6% (mean) and 50% (mode) within the ROPE region. According to criteria from Jeffreys (1961), BF_{10} values for the hypothesis of large correlations between tasks indicate moderate evidence ($BF > 3$) in favour of H_1 for the mean and median, anecdotal evidence for the mode, extremely strong evidence in favour of H_0 for the SD and MAD, and very strong evidence in favour of H_0 for the IQR.

Overall, these results indicate moderate correlations between tasks, such that for each RT index (i.e., mean, median, mode, SD, MAD, IQR), increases in a given index for the simple version of the SRT are associated with increases in the same RT index for the gamified SRT (DoggoNogo). However, H_1 can only be accepted for correlations between tasks' mean and median RTs, while H_1 is rejected across all indices of dispersion, and there is insufficient evidence to draw conclusions on Mode RT. These associations are weaker than we would expect of tasks measuring the same construct, with median values for indices of centrality only on the border of our threshold, and therefore insufficient to confirm good convergent validity.

Figure 4

Posterior distributions for correlations between tasks' RT indices.



Note. Blue regions represent the region of practical equivalence ($ROPE \leq 0.5$)

Probabilities of different effect sizes as categorised by Cohen (1988) can be seen in Table 2.

Table 2

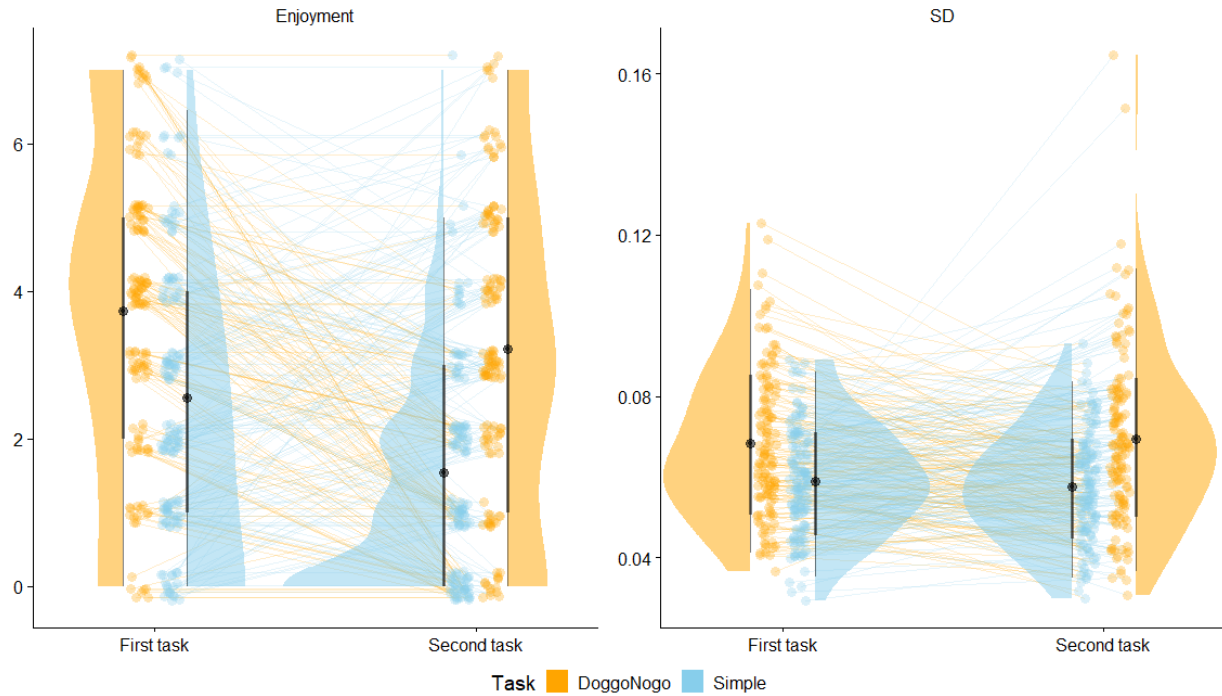
Probabilities for correlation effect sizes

	Small	Moderate	Large
Mean	0.000	0.088	0.912
Median	0.000	0.051	0.903
Mode	0.000	0.478	0.522
SD	0.020	0.965	0.015
MAD	0.019	0.958	0.023
IQR	0.018	0.957	0.025

In comparing the hypothesis that there are differences in enjoyment ratings between the simple SRT and the gamified (DoggoNogo) version of the SRT (H_1), and the null hypothesis, that no such difference exists (H_0), we found that DoggoNogo received higher enjoyment scores ($M = 3.46$, $SD = 1.9$) than Simple ($M = 2.07$, $SD = 1.74$). The result, $\delta = 0.76$, 95% CI [0.59, 0.93], $BF_{10} > 1Q$ indicates extreme evidence in favour of H_1 which is sufficient to discard H_0 in favour of H_1 . The distribution of enjoyment ratings by task can be seen in Figure 5. Controlling for task order did not nullify the effect of task on enjoyment (Appendix B).

Figure 5

Density and scatter plots showing differences in participant enjoyment ratings and RT variability across tasks and task order, with participants' values joined by lines coloured according to task order.



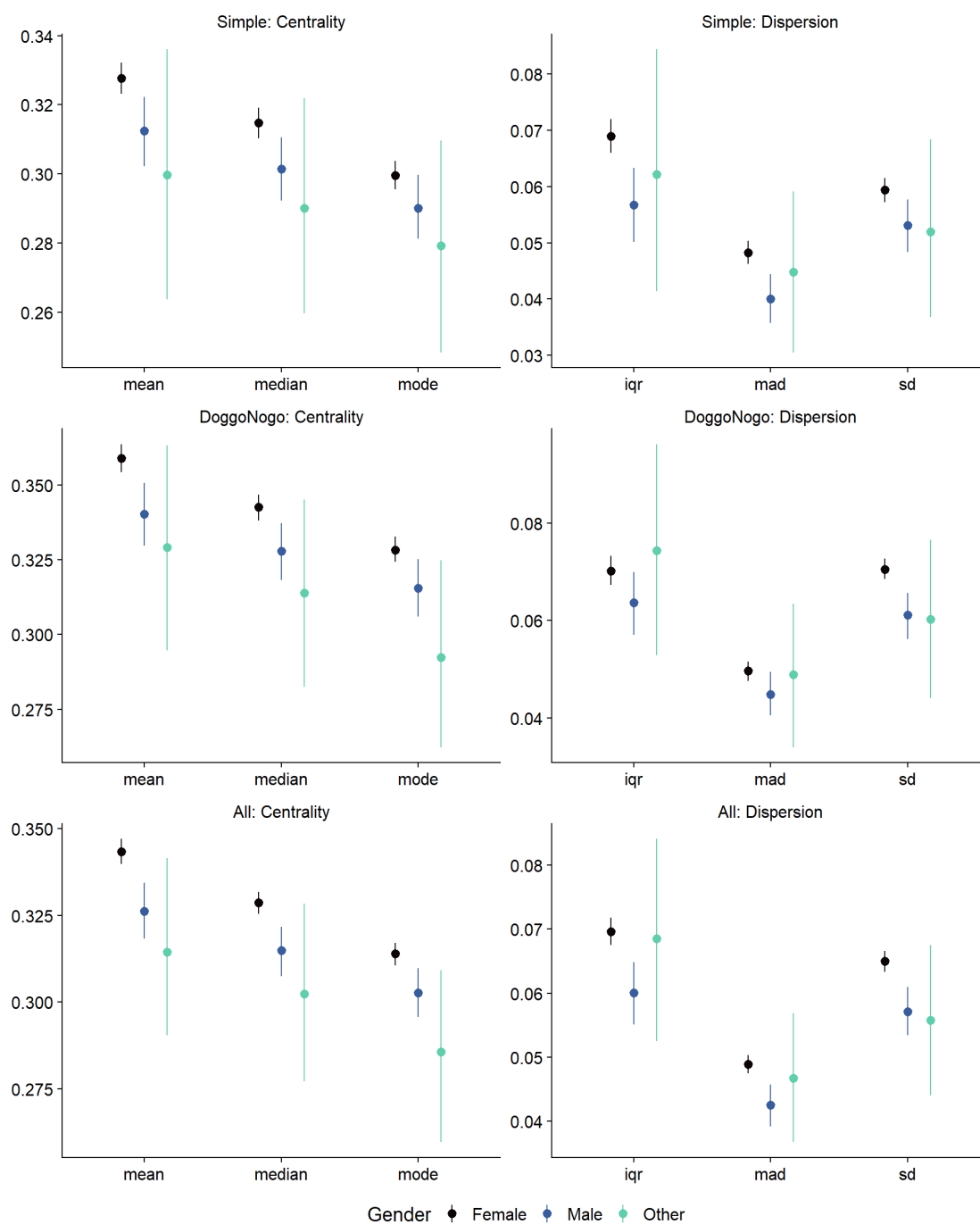
We next compared the hypothesis that there are differences in RT SDs between the simple SRT and the gamified (DoggoNogo) version of the SRT (H_1), and the null hypothesis, that no such difference exists (H_0). DoggoNogo received larger RT SDs ($M = 0.07$, $SD = 0.02$) than SRT ($M = 0.06$, $SD = 0.01$). The result, $\delta = 0.62$, 95% CI [0.46, 0.79], $BF_{10} = 14.2B$ indicates extreme evidence in favour of H_1 which is sufficient to discard H_0 in favour of H_1 . The distribution of SDs by task can be seen in Figure 5. The effect of task on RT variability held when controlling for task order (Appendix C).

Exploratory

Individual Differences. Full results of our hierarchical regressions predicting RT indices from Gender and Handedness (and their interactions with task version) can be found in Appendices D and E, respectively. Male and non-binary participants were faster than female participants across indices of centrality, with median reductions of between .011s and .017s for male participants and between .026 and .029 for non-binary participants (all $pd > .976$ indicating likely existence of an effect, [Makowski, Ben-Shachar, Chen, et al., 2019](#)). Parameters converged ($\hat{R} \leq 1.01$, [Vehtari et al., 2021](#)) and ESS values were sufficient (≥ 1000 , [Bürkner, 2017](#)) for all except non-binary mean RTs. These findings held for male but not non-binary participants when the interaction with task version was added to the models. A similar pattern is seen in dispersion indices, with male participants exhibiting lower RT variability than female participants (all $pd > .999$ indicating certain existence of an effect). These effects held for SD but not MAD or IQR, which dropped below $pd = .960$ ('possibly existing') when the interaction term was added. Parameters for non-binary participants did not converge. Overall, these results suggest that men have faster processing speeds than women, of approximately 10-20ms, irrespective of task. Marginal means with 95% CI (HDI) can be seen in Figure 6.

Figure 6.

RT indices by Gender across tasks



Note. Points represent means and error bars are constructed from credible intervals (HDI).

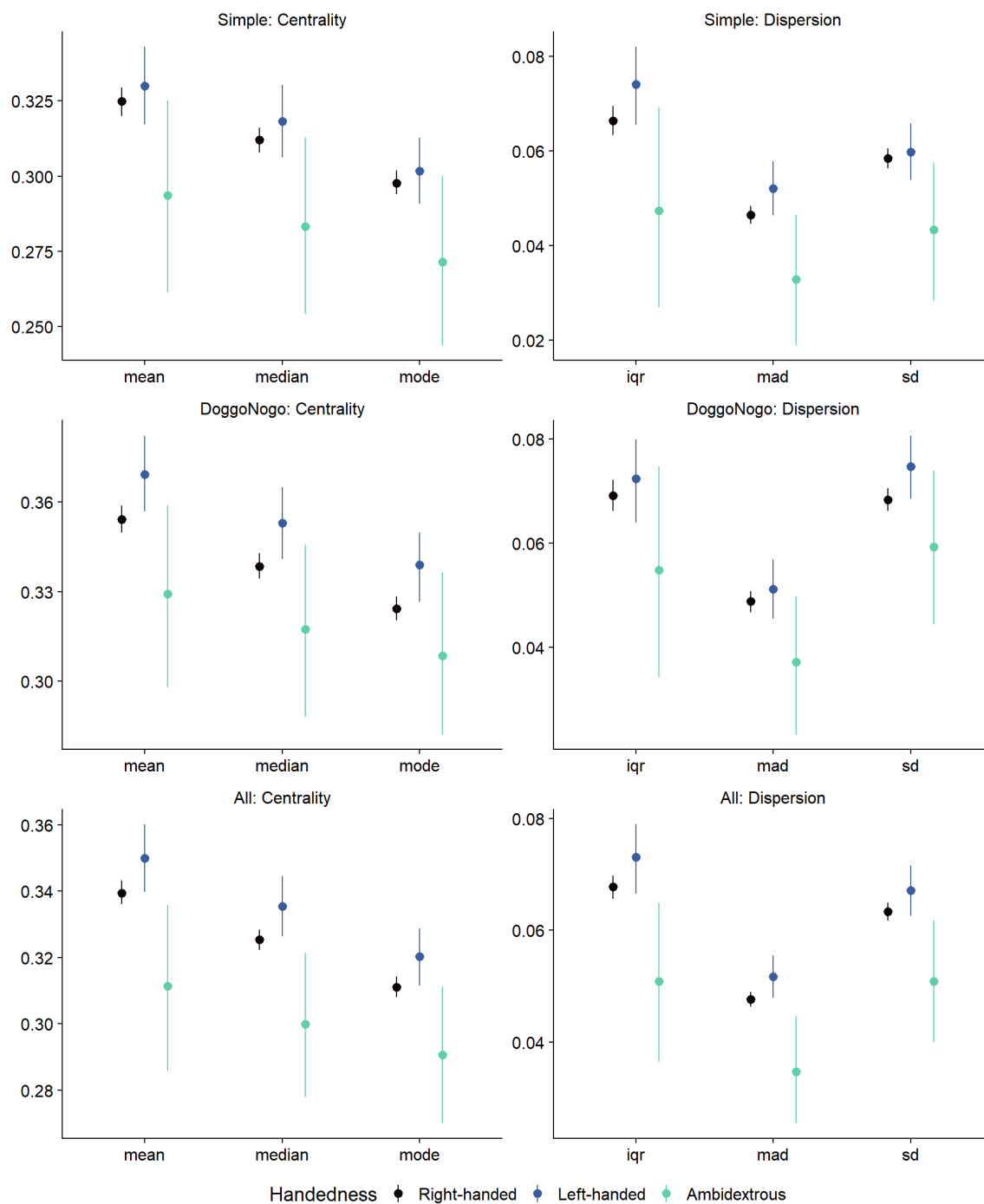
Across centrality models, addition of the interaction between gender and task significantly improved model fit ($\Delta R^2 = .15$ to $.16$), as well as for the SD model ($\Delta R^2 = .10$), but made little difference to the MAD and IQR models.

When assessing the effect of handedness on RT indices, we found that left-handed participants were significantly slower in terms of median and mode, but not mean RT, by about 10ms, while ambidextrous participants were faster across mean and median (but not mode) by 25-28ms. All parameters converged and ESS values were sufficient. Addition of interaction terms to these models consolidated findings concerning left-handed participants, with reduced speed across all indices of centrality by 14-15ms. In contrast, *pd* values for differences in ambidextrous individuals dropped below .95, suggesting non-significant effects, except for MAD, which had insufficient ESS. In terms of dispersion, left-handed participants had marginally larger MADs but this did not hold when interaction terms were added, with *pd* dropping below .95. Addition of the interaction revealed a likely effect of left-handedness on SD (*pd* = .974) irrespective of task. Marginal means with 95% CI (HDI) can be seen in Figure 7.

Across centrality models, inclusion of interactions between handedness and task improved model fit ($\Delta R^2 = .15$ to $.16$), as well as for the SD model ($\Delta R^2 = .09$), but did not improve MAD or IQR models.

Figure 7

RT indices by handedness across tasks



Note. Points represent means and error bars are constructed from credible intervals (HDI).

Optimization. Participant feedback included several comments on the repetitiveness and tediousness that is characteristic of simple reaction time tasks. With the aim of improving the efficiency of these tasks, without undermining data quality Figure 8 shows the convergence of RT indices with true RTs across trials. Cumulative RT indices converge rapidly for DoggoNogo, by around 40 trials, compared to the SRT which takes over 100 trials. We fitted polynomial (second- and third-degree) and general additive mixed models to RT data for each task and plotted predicted effects. As can be seen in Figure 9, predicted RTs are at their lowest in the ISI range 240-400ms across tasks and models.

Figure 8

Convergence of RT indices by trial number across tasks

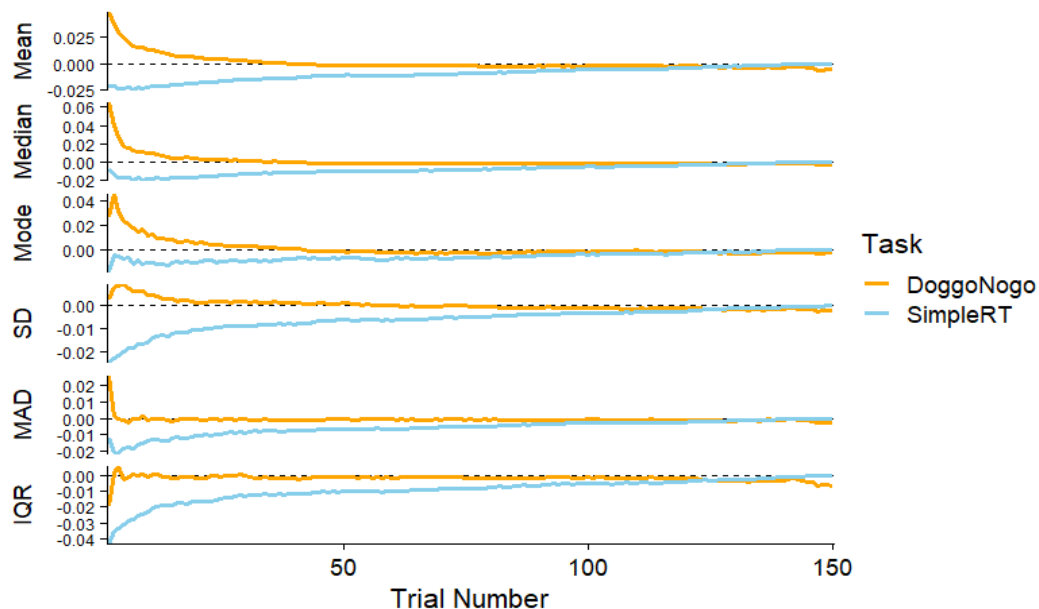
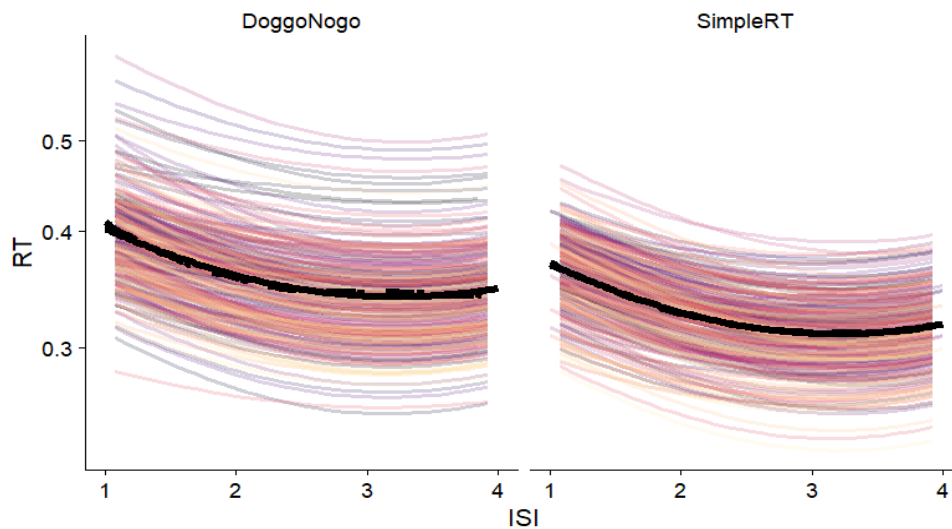


Figure 9

Predicted RTs across ISIs for each task



Note. Black lines represent predictions from second-degree polynomial (solid black line), third-degree polynomial (dotted black line) and generalized additive model (dashed black line).

Coloured lines represent participant-level predictions.

Discussion

Overall, participants found DoggoNogo, our gamified SRT incorporating a scoring system and implicit competition through performance dependent dynamic RT thresholds, significantly more enjoyable than the non-gamified SRT, irrespective of the order in which they completed the two tasks. This result is consistent with past research showing that participants prefer cognitive tasks when they have game-like elements ([Hamari et al., 2014](#); [Lumsden et al., 2016](#); [Robison & Nguyen, 2023](#)).

Our results are partially consistent with Robison and Nguyen ([2023](#)), who found significantly lower boredom in a task incorporating a competition element but no differences in

participants' boredom ratings between tasks with and without a points system. Although we did not isolate the effects of individual game components, our findings raise the possibility that implicit self-competition may be sufficient to increase enjoyment. This requires further investigation but would have important implications for task design when seeking to establish participants' optimal performance capacity.

Contrary to our expectations, participants' RT variability was greater during DoggoNogo, irrespective of task order. However, we cannot necessarily infer poorer engagement from this. Several participants indicated in written feedback⁴ that the busier background in DoggoNogo made the task more challenging. Because the visual clutter was unevenly distributed across the display region in which targets could appear, the extent to which the background impacted the ease of spotting the targets varied from trial-to-trial. Combined with evidence of greater enjoyment, it seems likely that this explains increased RT variability over and above any difference in engagement. This should be assessed using our task datasets, available on GitHub, which contain target coordinates (*Stimulus_X*, *Stimulus_Y*).

The difference in perceptual load between DoggoNogo and the SRT might also explain why correlations between tasks' RT indices were not higher, as would be expected in assessing for convergent validity. Pending assessment of the effect of stimulus coordinates, future iterations of the task may benefit from simplifying scenes to achieve more equivalent perceptual load across tasks (for example, replacing the background with an empty field, or restricting the

⁴ Participant feedback can be found at https://realitybending.github.io/DoggoNogoValidation/analysis/1_cleaning.html#experiment-feedback

region in which the bone may appear to the less visually cluttered area of the existing scene), making it easier to identify the target when it appears.

In line with previous research ([Ghisletta et al., 2017b](#); [Talboom et al., 2021](#)), we found that male participants had shorter RT indices of centrality than female participants, by around 10-20ms. This effect held across tasks, with a larger difference observed in the gamified task. Furthermore, males demonstrated reduced RT variability than females, but this only held for SD (not MAD or IQR) across tasks. These results are consistent with those found by [Ghisletta et al. \(2017a\)](#), who posited a role of sex hormones, however subsequent investigations do not support this ([Donadi et al., 2020](#); [Muley et al., 2023](#)). Future research should check for differences in experience in, for example, action games which might explain differences in motor speed; it would also be valuable to incorporate an initial speed assessment so that motor speed capacity can be disentangled from cognitive PS.

We found an effect of handedness, with right-handed participants demonstrating shorter RTs than left-handed participants across tasks. This may highlight a limitation of the construction of our tasks, which required participants respond with the arrow keys on the right of their keyboard; previous research has found that when participants respond with their dominant hand, there is no effect of handedness on SRT ([Woods et al., 2015](#)). Future adjustments to our tasks could therefore involve adapting response keys according to participant preference, giving the option of substituting arrows with, for example, ‘w’ (up), ‘a’ (left), ‘s’ (down) and ‘d’ (right) on a QWERTY keyboard.

Initial inspection of optimal trial numbers and ISI bounds indicate RTs are lowest for both tasks between 240 and 400ms that cumulative RTs converge with true RTs considerably

faster for DoggoNogo, requiring fewer than 50 trials to obtain a reliable index, while the non-gamified version required over double the number of trials. This demonstrates that, despite increased RT variability, our use of gamified features does improve efficiency of the simple reaction time task.

Some participants reported confusion and frustration at not knowing the parameters for successfully catching bones during DoggoNogo. A simple solution could be if bones moved towards doggo at speeds varying according to RT threshold, thereby ensuring coverage of regions with lower perceptual load across every trial. A more drastic option could be a side-scrolling format where obstacles move towards the participants' running character at different speeds (varying according to RT thresholds); participants must eliminate obstacles as quickly as possible before collision, earning points according to distance. In addition to helping participants to understand their changing RT thresholds and improving ecological validity, this would give more of a sense of continuity to the game. It would also offer an additional diagnostic tool by providing a secondary measure of RT (i.e., distance from character).

In conclusion, participants expressed significant preference for the gamified simple reaction time task, despite its increased demands. Though convergent validity with a non-gamified version was relatively low, with refinements to visual presentation, DoggoNogo shows promise as an engaging alternative to traditional PS tasks with optimised reliability and efficiency.

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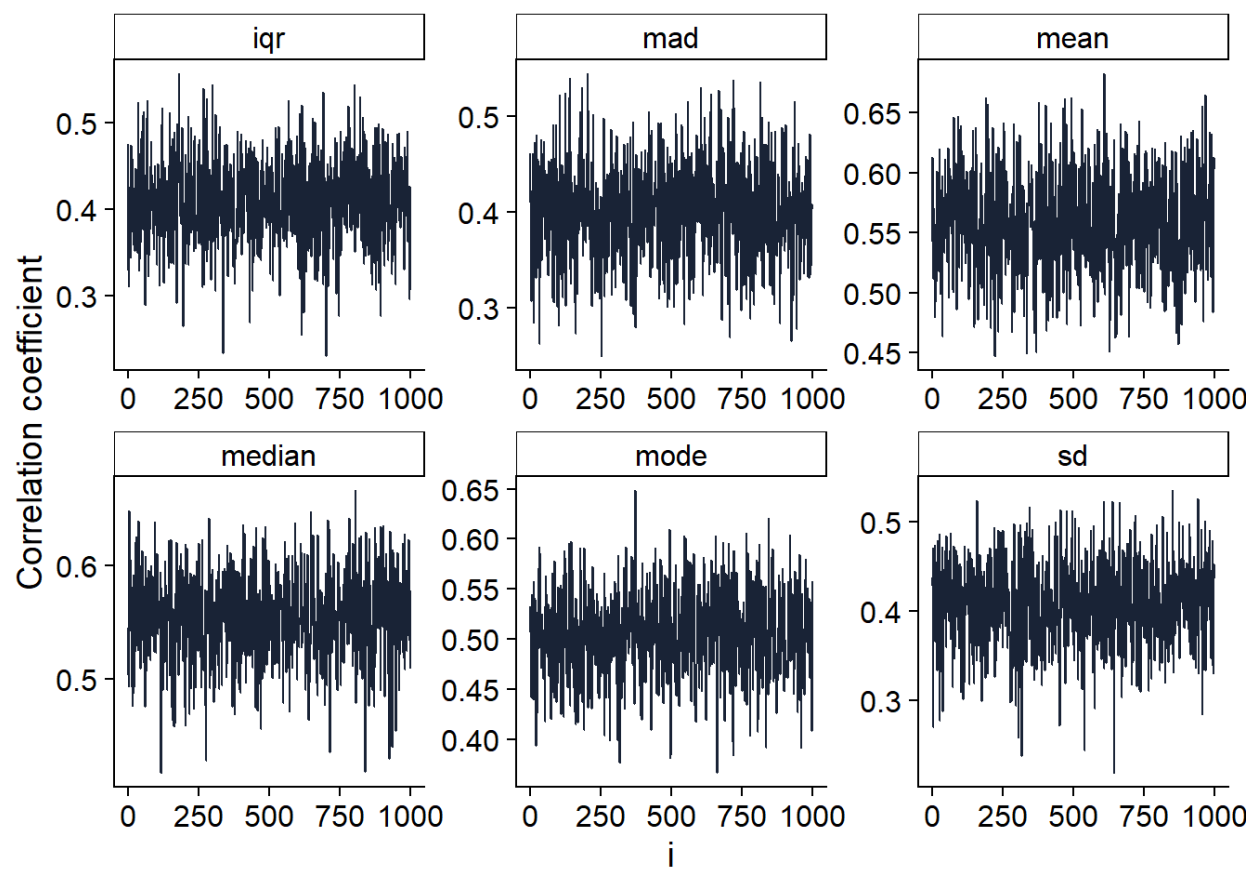
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Appendices

Appendix A

Trace plots for correlations between task RT indices



Appendix B

Predicting enjoyment from task (model 1) and controlling for task order (model 2)

Enjoyment ratings were rescaled as a proportion (0-1).

Parameter	Model 1					Model 2				
	Median	CI	pd	Rhat	ESS	Median	CI	pd	Rhat	ESS
Intercept	0.494	[0.464, 0.525]	1	0.998	2941.53	0.534	[0.491, 0.579]	1.00	1.000	1842.98
Simple	-0.199	[-0.240, -0.163]	1	0.998	4780.03	-0.315	[-0.363, -0.264]	1.00	1.000	2770.43
DoggoNogo:SimpleFirst						-0.075	[-0.141, -0.020]	0.99	0.999	1903.16
Simple:SimpleFirst						0.145	[0.090, 0.208]	1.00	1.000	1655.44
Sigma	0.222	[0.204, 0.241]	1	1.005	549.53	0.207	[0.192, 0.225]	1.00	1.006	649.29

Model performance indices

Model	ELPD	ELPD SE	LOOIC	LOOIC SE	WAIC	R2	R2 (marg.)	R2 (adj.)	R2 (adj. marg.)	ICC	RMSE	Sigma
1	-19.98	15.46	39.97	30.91	30.67	0.37	0.13	0.19	0.12	0.27	0.20	0.22
2	3.53	15.54	-7.06	31.07	-21.52	0.45	0.18	0.26	0.16		0.18	0.21

Model comparison

ELPD difference (Model 1) = -23.5; SE difference = 6.5

Appendix C

Predicting RT variability (SD) from task (model 1) and controlling for task order (model 2)

Parameter	<u>Model 1</u>					<u>Model 2</u>				
	Median	CI	pd	Rhat	ESS	Median	CI	pd	Rhat	ESS
Intercept	0.069	[0.067, 0.071]	1	1.00	1495.35	0.068	[0.065, 0.071]	1.00	1.000	1295.05
TaskSRT	-0.011	[-0.013, -0.009]	1	1.00	1685.47	-0.011	[-0.014, -0.007]	1.00	1.000	1896.65
DoggoNogo:SimpleFirst						0.001	[-0.003, 0.005]	0.69	0.999	1462.81
SRT:SimpleFirst						0.001	[-0.003, 0.005]	0.77	1.000	1430.41
sigma	0.013	[0.012, 0.014]	1	1.00	866.99	0.013	[0.012, 0.014]	1.00	1.002	578.35

Indices of model performance

Model	ELPD	ELPD SE	LOOIC	LOOIC SE	WAIC	R2	R2 (marg.)	R2 (adj.)	R2 (adj. marg.)	ICC	RMSE	Sigma
1	1509.99	26.91	-3019.99	53.81	-3037.25	0.44	0.09	0.23	0.08	0.38	0.01	0.01
2	1510.25	26.09	-3020.49	52.19	-3034.24	0.44	0.10	0.23	0.07		0.01	0.01

Model comparison

ELPD difference (model 1): -0.03; SE difference: 1.5

Appendix D

Parameters, performance and marginal means for RT indices predicted by Gender (model 1) and its interaction with task (model 2)

		Model 1				Model 2					
Model	Parameter	Median	CI	pd	Rhat	ESS	Median	CI	pd	Rhat	ESS
Mean	Intercept	0.343	[0.340, 0.347]	100.00	0.999	2409.750	0.359	[0.354, 0.364]	100.00	0.999	2372.289
	Male	-0.017	[-0.026, -0.008]	100.00	1.001	1622.286	-0.019	[-0.030, -0.008]	99.85	1.001	1424.575
	Other	-0.029	[-0.052, -0.001]	97.95	0.999	984.974	-0.030	[-0.066, 0.003]	95.20	1.003	1210.471
	Female:tasksrt						-0.031	[-0.037, -0.024]	100.00	0.999	2560.667
	Male:tasksrt						-0.028	[-0.042, -0.013]	100.00	1.001	1417.741
	Other:tasksrt						-0.030	[-0.080, 0.019]	87.25	1.002	1195.081
	Sigma	0.038	[0.036, 0.041]	100.00	1.003	1005.814	0.035	[0.033, 0.037]	100.00	1.001	1327.139
Median	Intercept	0.329	[0.325, 0.332]	100.00	1.001	2390.890	0.343	[0.338, 0.347]	100.00	1.001	2176.584
	Male	-0.014	[-0.021, -0.006]	100.00	0.999	1816.769	-0.015	[-0.026, -0.005]	99.80	1.001	1600.646
	Other	-0.026	[-0.051, -0.000]	97.65	1.002	1088.145	-0.029	[-0.059, 0.004]	96.25	1.002	1024.963
	Female:tasksrt						-0.028	[-0.034, -0.022]	100.00	1.000	2653.441
	Male:tasksrt						-0.026	[-0.040, -0.013]	100.00	1.003	1519.723
	Other:tasksrt						-0.024	[-0.066, 0.022]	83.75	1.005	973.539
	Sigma	0.036	[0.034, 0.038]	100.00	1.007	1079.042	0.033	[0.031, 0.035]	100.00	1.000	1356.709
Mode	Intercept	0.314	[0.311, 0.317]	100.00	0.999	2162.761	0.328	[0.324, 0.332]	100.00	1.000	2134.682

Validation of a Gamified Measure of Processing Speed

Model	Parameter	Model 1				Model 2			
		Median CI	pd	Rhat	ESS	Median CI	pd	Rhat	ESS
SD	Male	-0.011 [-0.019, -0.003]	99.65	1.000	2007.256	-0.013 [-0.023, -0.002]	99.20	1.000	1607.953
	Other	-0.028 [-0.051, -0.001]	99.00	1.003	1213.732	-0.036 [-0.068, -0.005]	98.40	1.003	939.977
	Female:tasksrt					-0.029 [-0.034, -0.022]	100.00	0.999	2041.821
	Male:tasksrt					-0.025 [-0.038, -0.012]	99.95	0.999	1452.569
	Other:tasksrt					-0.013 [-0.057, 0.032]	72.40	1.002	893.722
	Sigma	0.035 [0.033, 0.037]	100.00	1.009	1111.188	0.032 [0.030, 0.034]	100.00	1.002	1200.190
	Intercept	0.065 [0.063, 0.066]	100.00	1.000	2297.943	0.071 [0.068, 0.073]	100.00	1.000	2371.860
	Male	-0.008 [-0.012, -0.004]	100.00	1.001	1896.041	-0.009 [-0.015, -0.004]	100.00	1.000	2053.096
	Other	-0.009 [-0.020, 0.003]	92.90	1.001	1066.283	-0.010 [-0.024, 0.008]	89.40	1.006	742.669
	Female:tasksrt					-0.011 [-0.014, -0.008]	100.00	0.999	2929.938
MAD	Male:tasksrt					-0.008 [-0.015, -0.001]	98.75	1.000	1749.077
	Other:tasksrt					-0.009 [-0.032, 0.014]	75.60	1.006	709.254
	Sigma	0.017 [0.016, 0.018]	100.00	1.007	563.836	0.016 [0.015, 0.017]	100.00	1.004	989.138
	Intercept	0.049 [0.048, 0.050]	100.00	1.002	2268.171	0.050 [0.048, 0.052]	100.00	0.998	2252.880
	Male	-0.006 [-0.010, -0.003]	100.00	1.000	1862.471	-0.005 [-0.009, 0.000]	97.40	1.000	1736.196
	Other	-0.002 [-0.013, 0.008]	67.15	1.002	942.430	-0.001 [-0.016, 0.014]	54.45	1.004	782.004
	Female:tasksrt					-0.001 [-0.004, 0.001]	82.85	1.000	2383.600

Validation of a Gamified Measure of Processing Speed

		Model 1				Model 2					
Model	Parameter	Median	CI	pd	Rhat	ESS	Median	CI	pd	Rhat	ESS
IQR	Male:tasksrt						-0.005	[-0.012, 0.001]	94.15	0.999	1779.203
	Other:tasksrt						-0.004	[-0.025, 0.017]	64.65	1.001	826.890
	Sigma	0.015	[0.015, 0.016]	100.00	1.002	683.731	0.015	[0.015, 0.016]	100.00	1.003	677.099
	Intercept	0.070	[0.068, 0.072]	100.00	0.999	2159.433	0.070	[0.067, 0.073]	100.00	0.999	2090.487
	Male	-0.010	[-0.015, -0.005]	99.90	1.000	1850.013	-0.007	[-0.014, 0.000]	95.95	1.002	1391.538
	Other	-0.001	[-0.017, 0.014]	56.25	1.000	1353.208	0.004	[-0.016, 0.027]	64.45	1.001	1057.771
	Female:tasksrt						-0.001	[-0.005, 0.003]	71.30	1.000	2717.881
	Male:tasksrt						-0.007	[-0.016, 0.002]	94.10	1.001	1445.769
	Other:tasksrt						-0.012	[-0.045, 0.016]	77.50	1.004	1049.580
	Sigma	0.023	[0.022, 0.024]	100.00	1.001	858.321	0.023	[0.022, 0.025]	100.00	1.003	1281.063

Performance for Gender

Model	ELPD	ELPD_SE	LOOIC	LOOIC_SE	WAIC	R2	R2_adjusted	RMSE	Sigma
mean~Gender	1008.78	26.31	-2017.56	52.61	-2017.60	0.03	0.03	0.04	0.04
mean~Gender + task:Gender	1053.81	26.13	-2107.63	52.27	-2108.02	0.19	0.17	0.04	0.04
median~Gender	1052.02	26.13	-2104.05	52.27	-2104.10	0.03	0.02	0.04	0.04
median~Gender + task:Gender	1094.17	25.34	-2188.35	50.67	-2188.49	0.18	0.16	0.03	0.03

Validation of a Gamified Measure of Processing Speed

Model	ELPD	ELPD_SE	LOOIC	LOOIC_SE	WAIC	R2	R2_adjusted	RMSE	Sigma
mode~Gender	1065.64	26.67	-2131.29	53.34	-2131.33	0.02	0.01	0.03	0.03
mode~Gender + task:Gender	1112.03	26.05	-2224.07	52.09	-2224.19	0.18	0.17	0.03	0.03
sd~Gender	1447.96	28.39	-2895.92	56.79	-2895.96	0.03	0.02	0.02	0.02
sd~Gender + task:Gender	1471.96	28.14	-2943.93	56.28	-2944.45	0.13	0.10	0.02	0.02
mad~Gender	1511.31	34.16	-3022.62	68.33	-3022.32	0.03	0.02	0.02	0.02
mad~Gender + task:Gender	1509.53	33.52	-3019.07	67.03	-3019.13	0.04	0.01	0.02	0.02
iqr~Gender	1289.45	37.78	-2578.89	75.56	-2578.86	0.03	0.01	0.02	0.02
iqr~Gender + task:Gender	1288.08	36.59	-2576.16	73.18	-2576.63	0.03	0.00	0.02	0.02

Marginal means for Gender

Model	Task	Gender	Mean	95% CI	pd
mean~Gender		Female	0.343	[0.340, 0.347]	1
		Male	0.326	[0.318, 0.334]	1
		Other	0.314	[0.290, 0.341]	1
mean~Gender + task:Gender	DoggoNogo	Female	0.359	[0.354, 0.364]	1
		Male	0.340	[0.330, 0.351]	1
		Other	0.329	[0.295, 0.363]	1

Validation of a Gamified Measure of Processing Speed

Model	Task	Gender	Mean	95% CI	pd
median~Gender	Simple	Female	0.328	[0.323, 0.332]	1
		Male	0.312	[0.302, 0.322]	1
		Other	0.300	[0.264, 0.336]	1
	All	Female	0.329	[0.325, 0.332]	1
		Male	0.315	[0.307, 0.322]	1
		Other	0.302	[0.277, 0.328]	1
median~Gender + task:Gender	DoggoNogo	Female	0.343	[0.338, 0.347]	1
		Male	0.328	[0.318, 0.337]	1
		Other	0.314	[0.282, 0.345]	1
	Simple	Female	0.315	[0.310, 0.319]	1
		Male	0.301	[0.292, 0.311]	1
		Other	0.290	[0.260, 0.322]	1
mode~Gender	All	Female	0.314	[0.311, 0.317]	1
		Male	0.303	[0.296, 0.310]	1
		Other	0.286	[0.260, 0.309]	1
	DoggoNogo	Female	0.328	[0.324, 0.333]	1
		Male	0.316	[0.306, 0.325]	1

Validation of a Gamified Measure of Processing Speed

Model	Task	Gender	Mean	95% CI	pd
sd~Gender	Simple	Other	0.292	[0.262, 0.325]	1
		Female	0.300	[0.296, 0.304]	1
		Male	0.290	[0.281, 0.300]	1
	All	Other	0.279	[0.248, 0.310]	1
		Female	0.065	[0.063, 0.067]	1
		Male	0.057	[0.053, 0.061]	1
	DoggoNogo	Other	0.056	[0.044, 0.067]	1
		Female	0.071	[0.069, 0.073]	1
		Male	0.061	[0.056, 0.066]	1
sd~Gender + task:Gender	Simple	Other	0.060	[0.044, 0.077]	1
		Female	0.059	[0.057, 0.062]	1
		Male	0.053	[0.048, 0.058]	1
	All	Other	0.052	[0.037, 0.068]	1
		Female	0.049	[0.047, 0.050]	1
		Male	0.043	[0.039, 0.046]	1
	DoggoNogo	Other	0.047	[0.037, 0.057]	1
		Female	0.050	[0.048, 0.052]	1
		Male	0.043	[0.039, 0.046]	1
mad~Gender	Simple	Other	0.052	[0.037, 0.068]	1
		Female	0.049	[0.047, 0.050]	1
		Male	0.043	[0.039, 0.046]	1
	All	Other	0.047	[0.037, 0.057]	1
		Female	0.050	[0.048, 0.052]	1
		Male	0.043	[0.039, 0.046]	1
	DoggoNogo	Other	0.047	[0.037, 0.057]	1
		Female	0.050	[0.048, 0.052]	1
		Male	0.043	[0.039, 0.046]	1
mad~Gender + task:Gender	DoggoNogo	Female	0.050	[0.048, 0.052]	1

Validation of a Gamified Measure of Processing Speed

Model	Task	Gender	Mean	95% CI	pd
iqr~Gender	Simple	Male	0.045	[0.041, 0.049]	1
		Other	0.049	[0.034, 0.063]	1
		Female	0.048	[0.046, 0.050]	1
		Male	0.040	[0.036, 0.044]	1
		Other	0.045	[0.031, 0.059]	1
		Female	0.070	[0.067, 0.072]	1
	All	Male	0.060	[0.055, 0.065]	1
		Other	0.068	[0.053, 0.084]	1
		Female	0.070	[0.067, 0.073]	1
		Male	0.064	[0.057, 0.070]	1
iqr~Gender + task:Gender	DoggoNogo	Other	0.074	[0.053, 0.096]	1
		Female	0.069	[0.066, 0.072]	1
		Male	0.057	[0.050, 0.063]	1
	Simple	Other	0.062	[0.041, 0.084]	1
		Female	0.069	[0.066, 0.072]	1
		Male	0.057	[0.050, 0.063]	1

Appendix E

Parameters, performance and marginal means for RT indices predicted by Handedness (model 1) and its interaction with task (model 2)

Model	Parameter	Model 1				Model 2			
		Median CI	pd	Rhat	ESS	Median CI	pd	Rhat	ESS
Mean	Intercept	0.339 [0.336, 0.343]	100.00	1.001	2391.075	0.354 [0.350, 0.359]	100.00	1.000	2387.000
	LeftHanded	0.010 [-0.000, 0.021]	97.20	1.000	1689.296	0.015 [0.002, 0.029]	98.90	1.001	1473.080
	Ambidextrous	-0.028 [-0.055, -0.004]	98.95	1.003	1061.252	-0.025 [-0.057, 0.005]	94.55	1.002	1149.871
	RightHanded:tasksrt					-0.029 [-0.036, -0.023]	100.00	1.000	2905.675
	LeftHanded:tasksrt					-0.040 [-0.057, -0.020]	100.00	1.001	1494.738
	Ambidextrous:tasksrt					-0.036 [-0.077, 0.009]	94.00	1.001	1148.640
	Sigma	0.039 [0.037, 0.041]	100.00	1.003	1113.682	0.036 [0.034, 0.038]	100.00	1.000	1440.639
Median	Intercept	0.325 [0.322, 0.328]	100.00	1.001	2225.659	0.338 [0.335, 0.343]	100.00	0.999	2842.504
	LeftHanded	0.010 [0.000, 0.020]	97.75	1.002	1873.686	0.014 [0.002, 0.028]	98.60	1.002	1432.250
	Ambidextrous	-0.025 [-0.047, -0.002]	98.90	1.001	1062.864	-0.021 [-0.051, 0.006]	92.15	1.002	1162.734
	RightHanded:tasksrt					-0.026 [-0.032, -0.020]	100.00	0.999	2733.321
	LeftHanded:tasksrt					-0.035 [-0.051, -0.017]	100.00	1.002	1310.357
	Ambidextrous:tasksrt					-0.033 [-0.074, 0.007]	94.55	1.000	1346.352
	Sigma	0.036 [0.034, 0.038]	100.00	1.001	1035.529	0.033 [0.031, 0.035]	100.00	1.000	1251.701

Validation of a Gamified Measure of Processing Speed

		Model 1					Model 2				
Model	Parameter	Median	CI	pd	Rhat	ESS	Median	CI	pd	Rhat	ESS
Mode	Intercept	0.311	[0.308, 0.314]	100.00	0.999	2235.801	0.324	[0.320, 0.328]	100.00	0.999	2163.937
	LeftHanded	0.009	[0.001, 0.019]	97.60	1.000	1633.532	0.015	[0.001, 0.026]	98.35	0.999	1333.833
	Ambidextrous	-0.021	[-0.041, 0.002]	97.05	1.003	1283.500	-0.016	[-0.041, 0.013]	84.30	1.001	1009.893
	RightHanded:tasksrt						-0.027	[-0.032, -0.021]	100.00	0.999	3081.760
	LeftHanded:tasksrt						-0.037	[-0.055, -0.022]	100.00	1.000	1278.090
	Ambidextrous:tasksrt						-0.038	[-0.077, -0.000]	97.55	1.002	1171.427
	Sigma	0.035	[0.033, 0.037]	100.00	1.005	1119.327	0.032	[0.030, 0.034]	100.00	1.001	1386.602
SD	Intercept	0.063	[0.062, 0.065]	100.00	0.999	2343.757	0.068	[0.066, 0.071]	100.00	1.000	2267.914
	LeftHanded	0.004	[-0.001, 0.009]	93.75	0.999	1964.788	0.006	[0.000, 0.013]	97.35	0.999	1480.892
	Ambidextrous	-0.012	[-0.023, -0.001]	98.95	1.001	1170.220	-0.009	[-0.024, 0.006]	89.35	1.002	1032.368
	RightHanded:tasksrt						-0.010	[-0.013, -0.007]	100.00	1.000	2512.075
	LeftHanded:tasksrt						-0.015	[-0.024, -0.007]	100.00	0.999	1550.328
	Ambidextrous:tasksrt						-0.016	[-0.036, 0.005]	93.65	1.004	955.239
	Sigma	0.017	[0.016, 0.019]	100.00	1.006	700.388	0.017	[0.016, 0.018]	100.00	1.000	1056.888
MAD	Intercept	0.048	[0.046, 0.049]	100.00	1.000	2263.370	0.049	[0.047, 0.051]	100.00	1.000	2225.631
	LeftHanded	0.004	[0.000, 0.008]	97.40	0.999	1793.526	0.002	[-0.004, 0.008]	76.45	1.000	1413.850
	Ambidextrous	-0.013	[-0.023, -0.003]	99.60	1.001	773.162	-0.012	[-0.025, 0.002]	96.00	1.002	957.408

Validation of a Gamified Measure of Processing Speed

Model	Parameter	Model 1				Model 2			
		Median CI	pd	Rhat	ESS	Median CI	pd	Rhat	ESS
IQR	RightHanded:tasksrt					-0.002 [-0.005, 0.000]	95.35	1.000	2716.947
	LeftHanded:tasksrt					0.001 [-0.007, 0.008]	58.70	1.001	1130.135
	Ambidextrous:tasksrt					-0.004 [-0.023, 0.016]	67.70	1.003	991.647
	Sigma	0.015 [0.015, 0.016]	100.00	1.005	755.985	0.015 [0.015, 0.016]	100.00	1.006	844.111
	Intercept	0.068 [0.066, 0.070]	100.00	1.000	2087.889	0.069 [0.066, 0.072]	100.00	1.000	2390.661
	LeftHanded	0.005 [-0.002, 0.011]	94.00	1.001	1886.467	0.003 [-0.006, 0.011]	75.25	1.001	1558.167
	Ambidextrous	-0.017 [-0.030, -0.002]	98.90	1.000	1507.747	-0.014 [-0.035, 0.005]	91.95	0.999	1145.688
	RightHanded:tasksrt					-0.003 [-0.007, 0.001]	90.35	0.999	2562.280
	LeftHanded:tasksrt					0.002 [-0.009, 0.013]	61.10	1.001	1546.263
	Ambidextrous:tasksrt					-0.007 [-0.036, 0.023]	68.50	1.000	1079.669
	Sigma	0.023 [0.022, 0.024]	100.00	1.003	902.344	0.023 [0.022, 0.024]	100.00	1.000	1163.518

Performance indices for handedness

Model	ELPD	ELPD_SE	LOOIC	LOOIC_SE	WAIC	R2	R2_adjusted	RMSE	Sigma
mean~Handedness	1004.42	25.02	-2008.85	50.03	-2008.86	0.02	0.01	0.04	0.04
mean~Handedness + task:Handedness	1049.75	24.32	-2099.49	48.64	-2099.58	0.18	0.16	0.04	0.04
median~Handedness	1049.37	25.31	-2098.73	50.61	-2098.75	0.02	0.01	0.04	0.04

Validation of a Gamified Measure of Processing Speed

Model	ELPD	ELPD_SE	LOOIC	LOOIC_SE	WAIC	R2	R2_adjusted	RMSE	Sigma
median~Handedness + task:Handedness	1091.39	24.42	-2182.78	48.84	-2182.93	0.17	0.15	0.03	0.03
mode~Handedness	1063.19	26.21	-2126.38	52.42	-2126.41	0.02	0.00	0.03	0.03
mode~Handedness + task:Handedness	1109.51	25.69	-2219.02	51.38	-2219.16	0.18	0.16	0.03	0.03
sd~Handedness	1443.64	26.55	-2887.29	53.10	-2887.32	0.02	0.00	0.02	0.02
sd~Handedness + task:Handedness	1468.33	25.71	-2936.65	51.42	-2936.78	0.11	0.09	0.02	0.02
mad~Handedness	1510.53	32.11	-3021.06	64.22	-3021.13	0.02	0.01	0.02	0.02
mad~Handedness + task:Handedness	1508.69	31.67	-3017.38	63.34	-3017.47	0.03	0.01	0.02	0.02
iqr~Handedness	1288.22	34.67	-2576.45	69.33	-2576.41	0.02	0.01	0.02	0.02
iqr~Handedness + task:Handedness	1286.45	33.99	-2572.90	67.97	-2573.03	0.03	0.00	0.02	0.02

Marginal means for handedness

Model	Task	Handedness	Mean	95% CI	pd
mean~Handedness	All	Right-handed	0.339	[0.336, 0.343]	1
		Left-handed	0.350	[0.340, 0.360]	1
		Ambidextrous	0.311	[0.286, 0.336]	1
mean~Handedness + task:Handedness	DoggoNogo	Right-handed	0.354	[0.350, 0.359]	1
		Left-handed	0.369	[0.357, 0.382]	1

Validation of a Gamified Measure of Processing Speed

Model	Task	Handedness	Mean	95% CI	pd
median~Handedness	Simple	Ambidextrous	0.329	[0.298, 0.359]	1
		Right-handed	0.325	[0.320, 0.329]	1
		Left-handed	0.330	[0.317, 0.343]	1
	All	Ambidextrous	0.294	[0.261, 0.325]	1
		Right-handed	0.325	[0.322, 0.328]	1
		Left-handed	0.335	[0.326, 0.345]	1
median~Handedness + task:Handedness	DoggoNogo	Ambidextrous	0.300	[0.278, 0.321]	1
		Right-handed	0.338	[0.334, 0.343]	1
		Left-handed	0.353	[0.341, 0.365]	1
	Simple	Ambidextrous	0.317	[0.288, 0.346]	1
		Right-handed	0.312	[0.308, 0.316]	1
		Left-handed	0.318	[0.306, 0.330]	1
mode~Handedness	All	Ambidextrous	0.283	[0.254, 0.313]	1
		Right-handed	0.311	[0.308, 0.314]	1
		Left-handed	0.320	[0.312, 0.329]	1
	DoggoNogo	Ambidextrous	0.291	[0.270, 0.311]	1
		Right-handed	0.324	[0.320, 0.328]	1
		Left-handed	0.320	[0.312, 0.329]	1

Validation of a Gamified Measure of Processing Speed

Model	Task	Handedness	Mean	95% CI	pd
sd~Handedness	Simple	Left-handed	0.339	[0.327, 0.350]	1
		Ambidextrous	0.309	[0.282, 0.336]	1
		Right-handed	0.298	[0.294, 0.302]	1
		Left-handed	0.302	[0.291, 0.313]	1
		Ambidextrous	0.272	[0.244, 0.300]	1
		Right-handed	0.063	[0.062, 0.065]	1
	All	Left-handed	0.067	[0.063, 0.072]	1
		Ambidextrous	0.051	[0.040, 0.062]	1
		Right-handed	0.068	[0.066, 0.071]	1
sd~Handedness + task:Handedness	DoggoNogo	Left-handed	0.075	[0.069, 0.081]	1
		Ambidextrous	0.059	[0.044, 0.074]	1
		Right-handed	0.058	[0.056, 0.060]	1
	Simple	Left-handed	0.060	[0.054, 0.066]	1
		Ambidextrous	0.043	[0.028, 0.058]	1
		Right-handed	0.048	[0.046, 0.049]	1
	All	Left-handed	0.052	[0.048, 0.056]	1
		Ambidextrous	0.035	[0.026, 0.045]	1
		Right-handed	0.048	[0.046, 0.049]	1
mad~Handedness	All	Right-handed	0.048	[0.046, 0.049]	1
		Left-handed	0.052	[0.048, 0.056]	1
		Ambidextrous	0.035	[0.026, 0.045]	1

Validation of a Gamified Measure of Processing Speed

Model	Task	Handedness	Mean	95% CI	pd
mad~Handedness + task:Handedness	DoggoNogo	Right-handed	0.049	[0.047, 0.051]	1
		Left-handed	0.051	[0.045, 0.057]	1
		Ambidextrous	0.037	[0.023, 0.050]	1
	Simple	Right-handed	0.047	[0.045, 0.048]	1
		Left-handed	0.052	[0.046, 0.058]	1
		Ambidextrous	0.033	[0.019, 0.046]	1
iqr~Handedness	All	Right-handed	0.068	[0.066, 0.070]	1
		Left-handed	0.073	[0.067, 0.079]	1
		Ambidextrous	0.051	[0.036, 0.065]	1
iqr~Handedness + task:Handedness	DoggoNogo	Right-handed	0.069	[0.066, 0.072]	1
		Left-handed	0.072	[0.064, 0.080]	1
		Ambidextrous	0.055	[0.034, 0.075]	1
	Simple	Right-handed	0.066	[0.063, 0.069]	1
		Left-handed	0.074	[0.065, 0.082]	1
		Ambidextrous	0.047	[0.027, 0.069]	1